

Issues in large-vocabulary interactive speech systems

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Interactive telephone access to large application databases demands reliable large-vocabulary speech recognition. Current speech recognition performance is limited, but may be boosted by the inclusion of database and application information in the recognition process. The use of corroborative alphanumeric information such as spelling can also give a considerable boost to application performance. Accurate models for the relationships between database contents, the vocabulary used to describe these items, and the spelling and pronunciation of this vocabulary are very important. When these are packaged with careful dialogue design, they can make large-vocabulary applications feasible. An example corporate telephone directory system is presented illustrating all of the above features.

1. Introduction

Telephone-based interactive voice response (IVR) systems have been available for some time in such areas as call control, voice mail, and simple information retrieval tasks. In many cases they have been hugely successful in reducing the cost of providing telephone services and have consequently enabled a number of new services. Such speech systems are usually based upon DTMF (dual tone multifrequency) telephone keypad entry. This provides the caller with the ability to either perform a menu-style navigation through a structured speech dialogue, or to enter numeric information, such as telephone numbers or account numbers. Recorded speech phrases are usually used as the method of prompting the caller or for presenting information to them.

Recent developments in speech recognition technology have facilitated the introduction of small vocabulary (tens of words) speech recognition into network based services. The recently launched BT CallMinder™ product, a network based answering machine, is an example of the effective use of this. It incorporates 'yes/no', isolated digit and keyword recognition into a highly usable service. This replaces the need for callers to remember DTMF key sequences, which can be daunting for some customers. Many potential applications, however, require even more versatility in their caller interface. The entry of names and addresses, for example, is difficult with DTMF requiring that telephone keypads are labelled with the letters of the alphabet — each key representing a group of letters. This is inherently an ambiguous form of entry, and many customers in the United Kingdom do not have alphabetic labels on their telephones.

Recognition technology is maturing quickly bringing improvements in the accuracy and flexibility of large-

vocabulary speech recognition. This, coupled with the continued trend towards faster, cheaper computing platforms, has made the use of large-vocabulary speech recognition in interactive network services a real possibility.

By developing an automatic corporate directory enquiry system giving access to thousands of telephone numbers, BT Laboratories have demonstrated that the access of databases over the telephone using large-vocabulary speech recognition is a viable possibility.

This paper discusses the issues that surround the development of such systems, and describes techniques which address these issues. The structure of the paper is as follows:

- section 2 presents a practical architecture for large-vocabulary IVR systems,
- section 3 gives an outline of the corporate directory enquiry system,
- section 4 discusses dialogue design and the human factors of such systems,
- section 5 discusses vocabulary design and the modelling of data relationships,
- section 6 discusses current recognition technology performance, and its limitations,
- section 7 discusses the techniques for maximising this recognition performance,
- section 8 discusses the engineering of large-vocabulary database access systems.

2. Large-vocabulary IVR system architecture

Figure 1(a) shows a system architecture for a traditional IVR system which needs to access an application database.

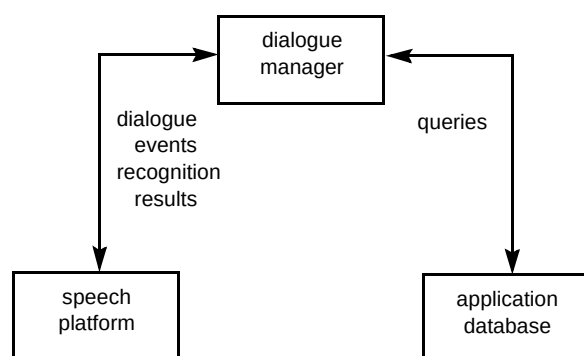


Fig 1(a) Traditional IVR architecture.

The 'dialogue manager' component defines the interface as seen by the caller. It is responsible for the sequencing of dialogue events, such as the playing out of prompts, and initiating speech recognition. In simple IVR applications the dialogue manager makes database searches directly using the results from DTMF entry or simple speech recognition.

The 'speech platform' component provides the speech functions required by the dialogue manager. It contains all of the speech recognition algorithms, DTMF detection, speech recording, dialogue prompt playback, and a telephony interface.

The 'application database' component defines all of the relationships between pieces of information used in the application. Commercially available database engines may be used for the application database component.

As this paper will show, it is desirable for systems requiring large-vocabulary speech access to databases to shield the dialogue manager from the need for direct application database interaction. This is because tight coupling is often required between the database and the speech recognisers in order to constrain the recognition vocabularies to the most likely anticipated responses.

In order to abstract this function away from the dialogue manager, a new 'information manager' component may be added to the architecture as shown in Fig 1(b). This component is responsible for mediating high level strategic requests from the dialogue manager and converting them into tactical actions. This will involve maintaining alternative hypotheses, querying the database for speech recognition constraints, and preparing speech recognisers with these constraints. The information manager will prepare

information such as confirmation items for the dialogue manager to offer to the caller. It will also ensure that vocabulary aliases are correctly managed throughout all of these actions.

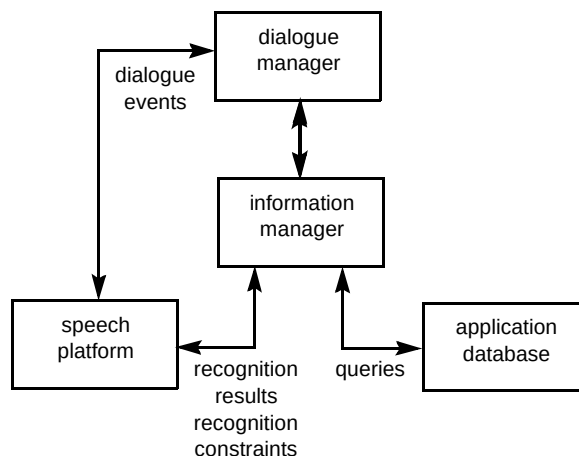


Fig 1(b) Large-vocabulary IVR architecture.

In large-vocabulary IVR systems additional speech platform features are required over and above those provided by traditional IVR platforms. These include:

- large-vocabulary, speaker-independent, sub-word-based speech recognition,
- dynamic recognition vocabulary modification and/or dynamic vocabulary generation in real time,
- text to speech.

Current large-vocabulary IVR development at BT Laboratories uses the ISAP speech platform [1] which meets all of these requirements.

3. Corporate directory enquiry system

BT Laboratories has produced a demonstration system providing access to a corporate telephone directory covering several thousand BT people from the Laboratories and the surrounding area.

The system provides prompt and accurate telephone number information for people, given only the first name and surname. It ensures that synonyms such as 'David' and 'Dave', or homophones such as 'Bayley' and 'Bailey', are considered to be equivalent during database access and in the subsequent confirmation and offering stages.

This system was built to demonstrate the feasibility of large-vocabulary IVR applications, and to provide a test bed on which to explore the issues involved. Emphasis was placed on producing a robust, usable service. Application

independence was also sought for as much of the system design as possible.

An example dialogue from this system is presented in Fig 2.

Welcome to the BT corporate directory enquiry system.	
Please say the surname	<i>Bailey</i>
And now, please SPELL the surname	<i>B-A-I-L-E-Y</i>
Please say the first name	<i>David</i>
Is the name Dave Bailey?	<i>Yes</i>
I have three entries matching that information,	
The number for Dave Bailey of	
Networks and Systems is 01473 64xxxx	
The number for David Bayley of Facilities	
and Technical Management is 01473 64xxxx	
The number for David Bailey of Business	
Operations is 01473 64xxxx	
I'll repeat that ..	

Fig 2 Example corporate directory system dialogue.

The issues that surround the development of such a system will be discussed in the following sections.

4. Dialogue design and human factors

The dialogue design of a large-vocabulary IVR application can have a dramatic impact on the performance of the system, given a particular underlying recognition performance [2], and consequently its acceptability to callers.

4.1 Choosing vocabularies

In order to keep a dialogue brief and reliable, it should start by asking for information that is accurately known by most callers. Callers must also know how to express this information when asked for it. Once all of the well-known information has been presented, the dialogue can proceed to ask for less well-known information, if this is needed to satisfy the enquiry. In these instances, a 'yes/no' question may be required to check whether a particular piece of information is known prior to asking for it.

The suitability of a particular type of information for input should also consider how many ways a given item or concept can be expressed. Vocabularies with a single form for each identifiable concept are called canonical, an example of such vocabularies being postcodes. Some vocabularies are very difficult to define, e.g. business areas in a corporate enquiry system, as separate callers may use a wide range of different vocabularies to describe the same item. It is very difficult to build accurate speech recognisers for these vocabularies.

Spelt or numeric vocabularies are generally highly canonical. If the information for that vocabulary is also well known for the given application, then it will probably provide a sound starting point on which to base an enquiry.

Examples of such systems include financial systems where knowing a personal account number is a reasonable prerequisite for using the system, or post-codes and spellings in applications requiring self-identification of the caller.

4.2 Message design for predictability

The design of the messages prompting the caller for desired information will itself affect the predictability of the response received. The use of clear, closed questions is sufficient to constrain many callers to present concise answers without extraneous speech. Early prototyping and small-scale trials of systems are essential in eliminating weaknesses in the dialogue design.

4.3 Mixed modality

There are several different modes of data entry available to the designer of a large-vocabulary IVR system. Among these are DTMF, large-vocabulary spoken recognition, large-vocabulary spelt recognition, and the use of 'yes/no' or DTMF response for menu selection. All of these may be mixed together in a single application, but is important to remain consistent in approach throughout the dialogue.

For example, in systems which mix spoken and spelt responses, experiments have shown that callers may be confused with dialogues, such as the one shown in Fig 3.

Please spell the surname	<i>F-O-S-T-E-R</i>
Now, please say the first name	<i>David</i>

Fig 3 Example from system using mix of spoken and spelt responses.

With this dialogue, callers may have a tendency not to listen closely to the prompts, and use the modality of a previous question to answer the current one. So, in the example in Fig 3, the first name may well be spelt rather than spoken.

If spelling is to be mixed with spoken responses, therefore, then it is advisable to always ask for the spoken word before asking for its spelling, even if the spoken response is never used by the system. Thus, the example in Fig 3 would become as shown in Fig 4.

Please say the surname	<i>Foster</i>
And now, please SPELL the surname	<i>F-O-S-T-E-R</i>
Now, please say the first name	<i>David</i>

Fig 4 Improved configuration for example of Fig 3.

Similar issues surround the mixing of DTMF with other modalities.

DTMF entry may be successfully used as a simultaneous alternative to speech input. In these instances, hybrid dialogue prompts such as 'Please say or use the keypad to enter the account number,' should be avoided. The prompt should only ask for a spoken response leaving DTMF as a fast-track option for experienced users who have found out about it from another source. BT's CallMinder is a successful example of this approach permitting the digits '1' and '2' to be used instead of 'yes' and 'no' if desired.

4.4 Confirmation and offering

It is often helpful to confirm the caller's request with them before offering the desired information, or service, related to the request. Such confirmation ensures that recognition errors are detected by the dialogue, and it also helps to orient the caller within the dialogue in preparation for the next phase.

Confirmation enables the caller and dialogue to proceed with confidence. It may not be necessary to confirm all of the information entered by the user, and it is acceptable to ask for further clarifying information after confirmation has occurred. Consider the corporate directory dialogue shown in Fig 5.

Please say the surname	<i>Foster</i>
And now, please SPELL the surname	<i>F-O-S-T-E-R</i>
Now, please say the first name	<i>David</i>
Please say yes or no,	
is the surname Foster?	Yes
I have a number of entries	
matching that information.	
Please answer yes or no,	
do you know the town name	Yes

Fig 5 Example of mid-query confirmation.

In the example of Fig 5, the caller is given confidence that the enquiry is proceeding well, and the system is now sure that the surname is 'Foster'. If the wrong surname is offered at this point, the caller is able to realise that the enquiry is in trouble prior to engaging in further lengthy dialogue to gather extra information.

Experience has shown that it can be confusing to expect callers to confirm information which they have not offered. For example, the dialogue shown in Fig 6 may be confusing to the caller.

Please say the surname	<i>Foster</i>
Please say yes or no,	
is the first name David?	Yes

Fig 6 Example which may cause confusion to the caller.

The exact words used for confirmation can be important. The choice of these words can come from two origins. Firstly, the spoken vocabulary, which was originally recognised, could be used directly. This has the advantage of confirming with the user the exact word which was spoken. Alternatively it may be necessary to confirm using portions of database entries which have matched the spoken information. This method has the disadvantage that if there are synonyms for the given vocabulary, the caller may be asked to confirm information using a synonym of the word they originally spoke rather than the word itself.

In the final stage of a dialogue, information is often offered to the caller in response to their enquiry, e.g. in the corporate directory example, a telephone number must be offered. The issues surrounding offering are similar to those of confirmation.

4.5 Distinguishability

It is sometimes desirable to confirm or offer items from a list of potential responses. In these instances, it is not wise to confirm lists of more than three items. The system must ensure that it does not try to draw distinctions between things which are indistinguishable to the user.

For example, consider the confirmation sub-dialogue in Fig 7.

Please say yes or no,	
is the name David Bailey?	No
Then is the name Dave Bayley?	

Fig 7 Example of an inappropriate confirmation sub-dialogue.

In the instance shown in Fig 7, it is inappropriate to expect the caller to draw a distinction between the two surname homophones 'Bayley' and 'Bailey'. It might also be inappropriate to expect the caller to distinguish between the synonyms 'David' and 'Dave'.

In order to avoid this, systems should test whether spoken vocabulary items share common pronunciations or have potential synonym confusions.

If an item is positively confirmed by the caller, all other items considered indistinguishable from that item should also be considered confirmed. This will mean the addition of homophones, and possibly synonyms also, prior to any re-searching the database using the confirmed information.

4.6 Speech generation

Large-vocabulary speech output may be handled in a few different ways. One simple way to achieve it is to make

recordings of each vocabulary item that may need to be spoken out by the dialogue. This is time consuming, and difficult to maintain, but it can produce a very high quality interface. This approach is suited to applications whose vocabularies are small to medium sized, and change slowly, if at all. One example of this may be a railway timetable enquiry system where the vocabulary of towns with stations changes very slowly. These towns, and the component words of dates and times may be recorded. Careful concatenation of these recordings in different contexts is able to provide a high quality interface.

For larger vocabularies, text to speech (TTS) is the only viable alternative. The designer must then choose whether all of the dialogue prompts are to be TTS generated, or simply the variable fields. Many recent TTS systems, including BT's Laureate, can be based on a particular individual's voice. This enables TTS generated variable fields to be spliced into recorded sentences from the same speaker. Care must then be taken with the prosody and intonation of the resulting sentences.

5. Vocabulary design and data modelling

5.1 The many forms of a word

As has been seen already, a database entry may be referred to using many different forms. Often, the caller cannot be sure that the word or spelling that they know for a certain piece of information is the form which found its way into the database.

Figure 8 shows a generalised entity relationship diagram relating together the different representations used for items in large-vocabulary systems. These representations are shown as 'spoken vocabulary', 'spelling vocabulary', 'pronunciations', and 'database representation'.

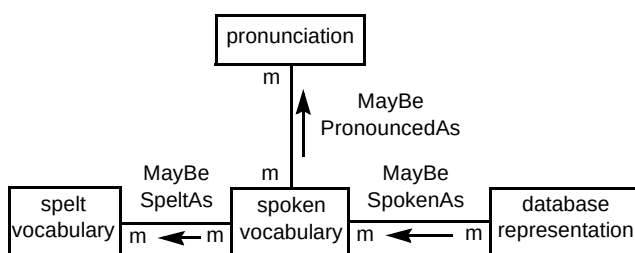


Fig 8 ERE diagram relating different representations of vocabulary items.

The database representation is the form in which an item is stored in the database. These database representations may be referred to using many different spoken words for a given item. Each spoken item has a spelling, but there may be more than one 'valid' spelling of an item in an application if callers are permitted to spell homophones or make common spelling errors. For this reason, spelt

vocabulary representations are considered to be separate from spoken representations.

Each word will have a pronunciation, but there may be more than one pronunciation for a given word (homonyms), and different words that share pronunciations (homophones).

5.2 Data management

Given the rich set of representations and relationships required by large-vocabulary applications, data preparation and management are important issues.

Figure 9 shows a typical data preparation process required to get from the source database text to the spoken, spelt and pronunciation forms as defined in Fig 8. Figure 9 only shows the preparation of the sets of valid tags for the entities of Fig 8. The process also implies the definition of the relations shown in Fig 8.

In text-based information systems, anomalies in the text stored in the database may often be compensated for by the use of re-keying of alternatives and visual browsing by the user. This is especially true of name and address information where the original data entry may have introduced spelling errors, or homophones of the correct words.

In large-vocabulary IVR applications, the information in the database must be associated with the vocabulary with which to describe it, and the valid pronunciations of this vocabulary. This requires the use of standardised synonym, abbreviation and pronunciation lexicons. Thus, text normalisation is often required on the raw database information in order to establish a regular set of database representations against which vocabularies and their relationships to the database may be prepared.

In the building of large-vocabulary recognisers from sub-word units, the quality of the pronunciation lexicons is often a significant factor affecting the recognition performance. As large-vocabulary IVR systems become more widespread, the management of corporate lexical information such as pronunciations and synonyms will become an increasingly important issue. The EU Onomastica project, which derived accurate pronunciations for many names, roads, towns and businesses in Europe attempted to anticipate this [3].

It is possible to set a balance between how much data preparation is performed off-line from the live application, and how much is performed dynamically. This is a design decision driven by the characteristics of the target application platform and the data processing requirements.

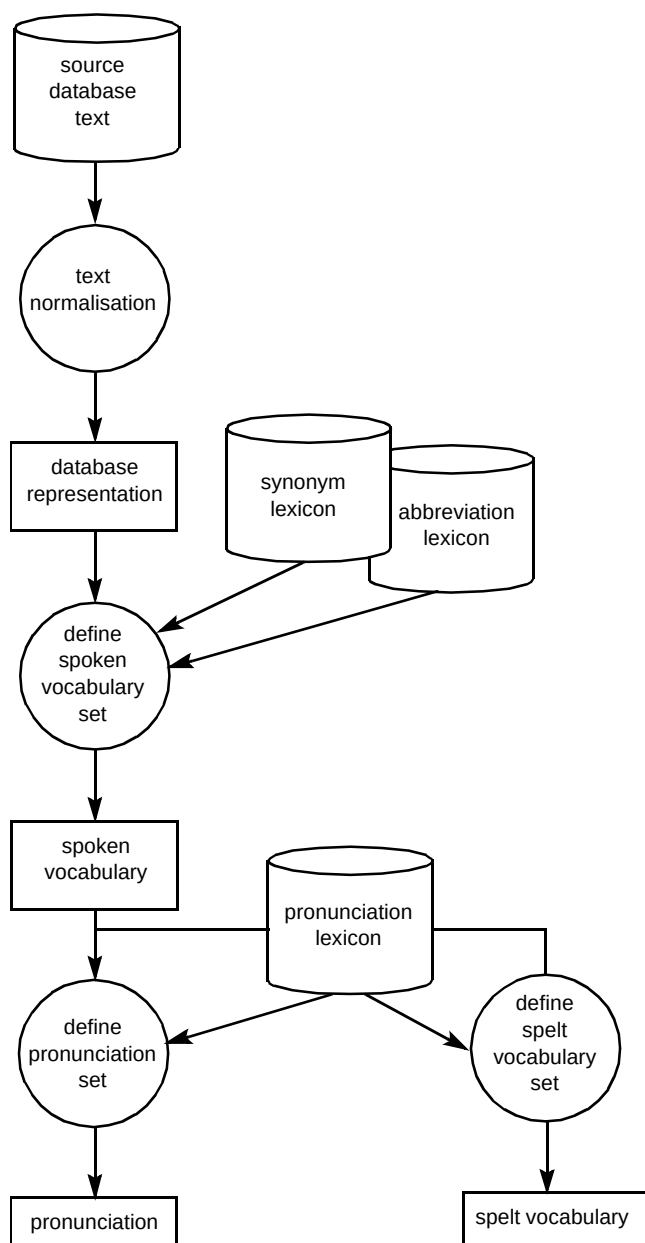


Fig 9 A typical data preparation process.

6. Recognition technology

Accurate speech recognition is essential if reliable services are to be offered. Large-vocabulary speech recognition performance is affected by several factors, including

- vocabulary — the number and similarity of words or phrases in the target vocabulary,
- training — whether the speaker has trained the recogniser prior to use,
- channel — effects of the channel (bandwidth, etc) and background noise,

- predictability — the predictability of response from the caller,
- pronunciations — the accuracy of the pronunciations upon which sub-word recognisers are based.

In network-based services with large vocabularies and an open caller group, all of these factors mitigate against accurate speech recognition. By way of example, dictation systems in which the user pauses between words can achieve a word accuracy in excess of 90% for vocabularies of tens of thousands of words. In order to achieve this, they must be trained by the particular user, use a high-quality microphone and use knowledge of the language being spoken to assist the recognition.

In contrast to this, speaker-independent, isolated-word recognition over the telephone network for some vocabularies will give comparable performance for vocabularies comprising only tens or hundreds of words. Fluent spelt and alphanumeric recognisers can work considerably more accurately over the network under the same conditions, giving comparable performance to above for vocabularies of thousands of words.

The need for careful engineering of systems with such performance is evident. It has been the approach of the authors to design for high application performance, attempting to prevent, hide, or minimise the impact of speech recognition errors. In order to achieve this, it is often helpful to use contextual constraints on the recognisers throughout all stages of the enquiry, and to maintain many alternative speech recognition hypotheses in the enquiry until corroborating information can be used to choose between them. The next section discusses these techniques in more detail.

7. Maintaining uncertainty and imposing constraints

7.1 The benefits of uncertainty

Many current speech recognisers do not simply return a 'best guess' candidate from a recognition, they can return several. This is termed the '*N*-best' list of candidates. Generally, each candidate word in this *N*-best list will have associated with it a score to indicate a confidence relative to the other words recognised.

The maximum number of candidates which may be returned will depend on the computational cost of keeping multiple word hypotheses active to the end of the recognition. It may also depend on certain rejection criteria in an attempt to return only credibly similar words from the recognition.

Where there are a number of words which are acoustically similar, it is likely that the correct word may be present in the *N*-best candidate words, but incorrectly

ranked. Figure 10 shows the typical performance trend of a best-1 recognition, and a best-5 recognition for a spoken large-vocabulary network-based speech recognition task.

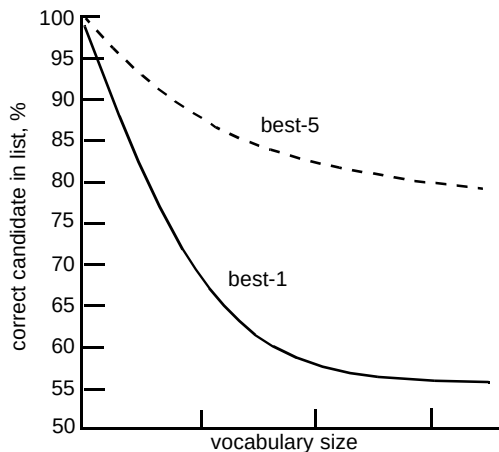


Fig 10 Best-1 and best-5 recognition performance versus vocabulary size for a typical large-vocabulary speech recognition task.

In some interactive speech systems it is not necessary to know the exact vocabulary item entered at the moment it is presented. This means that a number of uncertain candidates may be retained, and corroborative information may be used later to re-rank the candidates giving an opportunity for error recovery.

As a simple example, consider a directory access system. The caller is asked to state the first name, and then asked to state the surname. Two independent recognitions of these words may be performed, keeping the N -best candidate names from each recognition. A database search may then be performed finding all of the entries which match one of the first names in the list and one of the surnames in the list. It is probable that, due to redundancy in the database, only certain combinations of first names and surnames will be present in the result. If any matching pairs are located, then they may be rank ordered according to the scores of both of the candidates which matched, and presented to the caller for confirmation. In this simple example, a considerable application performance gain may be achieved over using the best-1 candidate alone for the search.

7.2 Constraining speech recognisers

The computational cost of large-vocabulary speech recognition often demands that the pruning of unlikely paths in the recognition network is performed throughout the matching of the word [4]. This can sometimes result in the correct candidate being pruned out early on in the recognition due to a poor match at the start of the utterance. It cannot then be recovered, even if the final recognition score would have been high.

In the example system described in the section above, this kind of error cannot be guarded against, and pruning errors will occur. This kind of effect is also noticeable in systems where acoustic fast-matches are used to reduce the search space of a recognition prior to a detailed acoustical match.

As already mentioned in section 6, the recognition performance is directly related to the degree to which one may predict, *a priori*, a caller's response to a question. If it is known that certain words are more likely to be said than others prior to an attempt to recognise them, then the recogniser can be weighted towards these words making it less likely that these words will be pruned out during the pattern match, and making them more likely to be recognised.

In some cases, as will be discussed in the next section, it is sensible to 'turn off' certain vocabulary items altogether. That is to say, inform the recogniser that there is no possibility that the particular word will be spoken. This mechanism is termed 'subsetting' throughout this paper. The benefit of vocabulary subsetting is that pattern matching effort is not wasted on improbable vocabulary items, and hence for a given computational resource, more effort can be channelled into recognising the word against a smaller set of reference models, more quickly, or with a higher accuracy.

With this technique, different weightings may still be applied to the remaining vocabulary items, termed 'soft subsetting', but this may not be desirable or possible with some speech recognisers, and a simple on/off constraint may be used, the surviving vocabulary items being treated as equally probable. This is termed 'hard subsetting'.

The use of *a priori* constraints on a pruning recogniser is a common approach in speech recognition, known generally as language modelling. The next section discusses the use of the database itself as a language model.

7.3 Database constraints

In large-vocabulary IVR systems, a language model may be formed dynamically from the contents of the database itself and applied as *a priori* constraints on the speech recognisers. These constraints are derived from previous recognition results, and the known database contents and are termed database constraints. This can provide a powerful method for reducing recognition pruning errors, reducing vocabulary sizes, and speeding up application performance.

The use of database constraints depends on the assumption that the caller is attempting to obtain an entry which is represented in the database. Figure 11 illustrates the use of database constraints by reconsidering the

corporate directory example discussed in the previous sections.

Consider the enquiry as a process of gradually closing down the number of possible database entries that the caller may be trying to access until a single entry may be selected with a certain degree of confidence. Initially, any one of the entries may be desired by the caller with equal probability. This hypothesis is labelled 'I' in Fig 11.

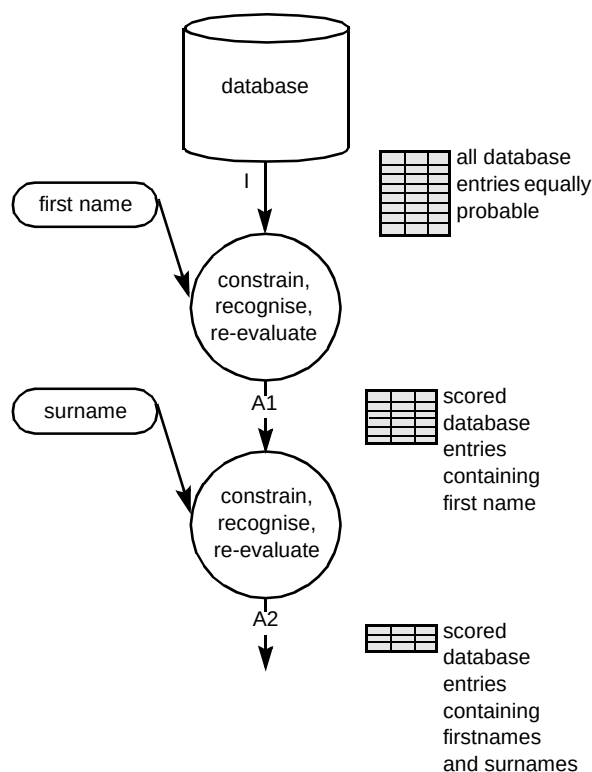


Fig 11 Using database constraints to improve application performance.

The first name recogniser is then constrained by this hypothesis, which in this case means that the recogniser uses all of the valid vocabulary represented in the database. So far, this is no different from the previous example. The first name is then recognised, and the database entry hypothesis is re-formulated to keep all entries which match one of the N -best first names, and discard those that do not. Of those that match, they are assigned relative scores depending on the recognition scores. This new hypothesis is labelled A1 in Fig 11.

Now, under the assumption that this recognition contained the correct result, the surname recogniser is then constrained to only recognise surnames which are still in A1. The surname is then recognised, and the database entries are re-scored again if they match one of the N -best surnames, or further entries are discarded if they do not. The resulting entry hypothesis is labelled A2.

A test may then be made on A2 to see whether the potential entries are now so few in number that the most probable of them may be confirmed with the caller. If there are too many still, then further information may be sought, for example a town or building name, etc.

This system cannot have worse performance than the previous example, and may well have better performance in terms of the proportion of correct entries that are offered. In some cases, this method can have dramatic performance gains. However, because of the assumptions on which it is based, this approach is sensitive to recognition errors in previous vocabularies, and it is also sensitive to callers presenting information which is not present in the database (e.g. the name of a colleague who has left the company). In either case, it may on occasion return database entries to the caller which bear no similarity to the original information. The following section discusses these weaknesses in more detail, and proposes mechanisms for avoiding them.

7.4 Alternative hypotheses and error recovery

The example presented in the previous section represented a single track approach to accessing the database. In that system it was assumed at all stages that:

- the previous recognition was correct,
- the caller presented information that was represented in the database.

In real systems, both of these assumptions have limited validity, and the application can be made much more robust to errors by attempting to detect when mistakes have been made and correct for them.

Consider the corporate directory system of Fig 12. The derivation of A2 is the same as described above. Assume however in this instance that the surname recogniser used to derive A2 indicated that an out-of-vocabulary item was detected. This may suggest that the caller presented a first name/surname pair that was not in the database, or it may indicate that an error was made recognising the first name. By re-recognising the surname without the first name constraint, a new estimate of the entries which are desired may be derived. This is shown as B1 in the diagram. There is now an alternative, possibly lower confidence, hypothesis track which the application may pursue. This may be explored in a similar manner to the previous track until it is either found to lead to the correct solution, or it too is found to be lacking.

This is an example of the use of an out-of-vocabulary rejection and an alternative lower confidence back-off strategy in an attempt to recover from an error. Alternatively multiple entry hypotheses may be compared

with one another to cross-check that the enquiry is proceeding well.

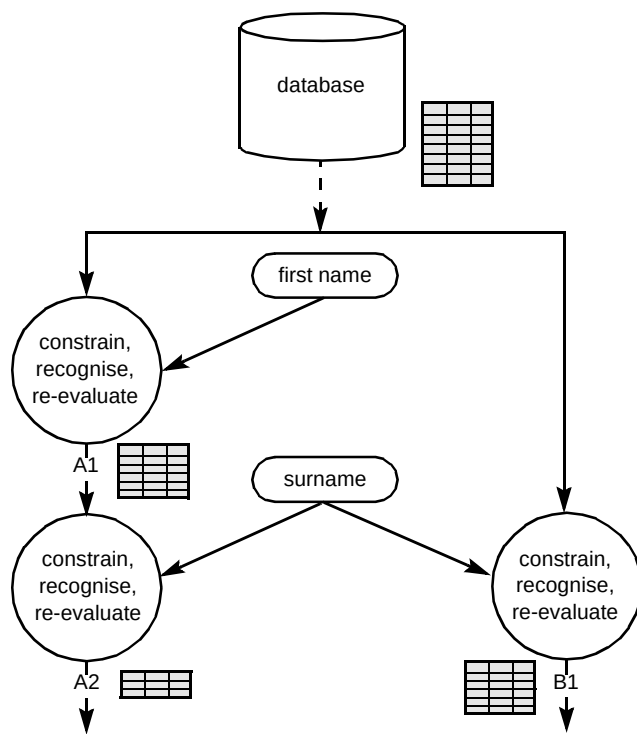


Fig 12 Using alternative hypotheses for error recovery.

Often, all of these approaches can be used together, with great effect, to provide error recovery and give indications of confidence.

7.5 Spelling and alphanumerics

As previously mentioned in section 6, the recognition of fluently spoken alphanumerics is generally more reliable for a given vocabulary size than spoken recognition. This fact, coupled with issues of canonical form discussed in section 4, mean that alphanumeric entry can provide a very reliable first-level constraint in some large-vocabulary IVR systems.

Examples of the effective use of alphanumerics include the customer account numbers, or the spelling of a surname. If the item is entirely numeric, for example, a telephone number, then DTMF may provide a totally reliable form of entry. Subsequent recognitions may be constrained by this information using a database constraint.

Of specific interest is the speaking and spelling of the same vocabulary item, for example a surname. Where this is acceptable to callers, using the techniques already discussed, combined spelt and spoken recognition may be made very reliable.

7.6 Application usage constraints

The application of constraints on the recognition need not be based on previous recognition results. They may be

based on any statistical knowledge that is available to the application designers.

For example, an application where this may prove useful is catalogue shopping where customers are identifying themselves by their own name and address. Of the tens of thousands of towns in the UK, the majority of callers may be concentrated in the top few cities and towns. The remaining towns being represented with decreasing frequency. Figure 13 shows a typical town vocabulary distribution that such an application may display.

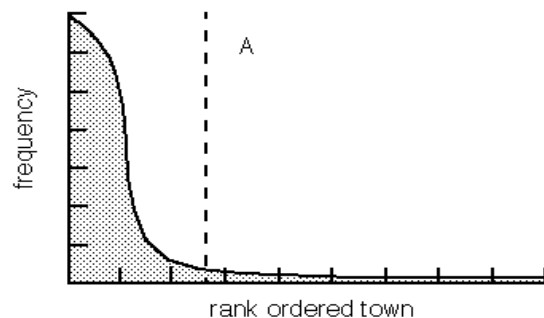


Fig 13 Typical usage statistics for geographical information systems.

This knowledge may be used in a similar way to other constraints. One approach may be to use a full vocabulary recogniser with weightings reflecting the sorts of distributions shown in Fig 13. If carefully balanced and tuned, such a system could provide improved average application performance while still enabling enquiries in the statistical tail of the graph to succeed.

An alternative approach is to deliberately exclude the unlikely vocabulary entries (i.e. use a soft or hard vocabulary subset), for example keeping only the towns to the left of line A on Fig 13. By excluding the tail of the distribution, the recognition performance on the top few likely words will be very reliable. Out-of-vocabulary rejection on such a small vocabulary may be reliable enough to detect when callers are presenting information in the tail of the distribution and react accordingly, for example asking for a corroborating spelling of the vocabulary item.

8. Engineering large-vocabulary database access systems

8.1 Database integration

The techniques of using database constraints, as discussed in section 7, may require high performance database engines to provide real time response. This is one limitation on the use of the technique, and presents serious engineering problems for multi-channel systems which are to interface into existing corporate databases.

It may be necessary to cache portions of the database, or to use looser derived statistical models of the database in order to get real time constraint performance.

Local data caching is realistic for read-only databases with slow update cycles, but more sophisticated approaches will be required for multi-channel access to fast-changing information sources, such as airline information systems.

8.2 Dynamic vocabulary modification

Section 7 discussed the benefits of vocabulary subsetting in speech recognisers. There are several different ways in which this can be implemented in practice, some of which are listed below:

- multiple recognition vocabularies, which have been pre-built for each expected context, are used — this approach quickly becomes unmanageable for even small systems,
- dynamic *a priori* probability modification of an existing large-vocabulary recognition vocabulary can be achieved in real time, and is a very effective approach for large vocabularies — Scahill [5] describes how BT speech recognisers implement this technique for finite state grammar recognisers (this can provide soft subsetting),
- dynamic vocabulary generation — using this method, recognition vocabularies are only built when they are required; this is currently a very effective technique for small vocabularies, and it is especially attractive for rapidly changing databases. This can provide hard subsetting, or soft subsetting when used in conjunction with dynamic *a priori* probability modification.

9. Conclusions

This paper has presented a number of techniques which enable some large-vocabulary IVR applications to become practical propositions with the current technology that is available.

These techniques are based on the philosophy that the speech recognition algorithms must have the maximum possible application information available to them prior to attempting recognition. The predictability of response from the caller is also a major influence on performance. Dialogue designs must be well considered and tested thoroughly.

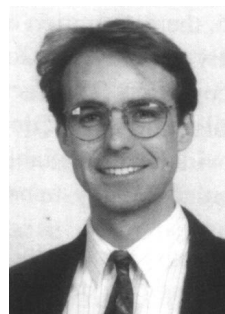
Such approaches will maximise the performance of an application for a given speech recognition accuracy.

However, some classes of application are still too large or undefined to be made sufficiently robust for use with current recognition algorithms. These will require fundamental advances in accuracy before they become viable.

The approaches presented have been tested by the development of a corporate directory system for BT Laboratories, providing reliable number information based on first name and surname alone for several thousand people. This system represents a major step forward, demonstrating that a whole new class of applications are now suitable for automation in the immediate future.

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